Xinyi Fang

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Research Interests

Agentic search; evidence-grounded multimodal reasoning; event understanding and forecasting; evaluation and adaptation of language and vision-language models. I am interested in how models select and synthesize information across sources to support reliable decisions, with applications to news analysis and AI for social good.

Education

Saint Louis University St. Louis, MO
Ph.D. in Computer Science *Sep. 2026 – Present*
TRACE AI; advised by Prof. Jin Zhao

Brandeis University Waltham, MA
Ph.D. Studies in Computer Science *Aug. 2025 – Sep. 2026*
Transferred to Saint Louis University to work with Prof. Jin Zhao

Worcester Polytechnic Institute Worcester, MA
Master of Science in Computer Science *Aug. 2023 – May 2025*
Thesis: *Enhancing Language Models for Classification on TAGs through Multi-Model Data Augmentation*
Adviser: Prof. Kyumin Lee

Guangzhou University Guangzhou, China
Bachelor of Science in Data Science *Aug. 2019 – May 2023*

Publications

1. **Xinyi Fang**, Jin Zhao. "How Hard Is It to Identify AI-Generated Meme Ideas?" *AACL-IJCNLP 2026 Student Research Workshop (SRW)*, to appear. [Code]
2. **Xinyi Fang**, Kyumin Lee, Yichuan Li. "LLM Profiling and Fine-Tuning with Limited Neighbor Information for Node Classification on Text-Attributed Graphs." *5th Workshop on KG and BigData (KGBigdata), IEEE Big Data 2025.* [Code]

Research Experience

Research Assistant, TRACE AI Sep. 2026 – Present
Saint Louis University; adviser: Prof. Jin Zhao *St. Louis, MO*

Cross-Document Event Graphs and Multimodal Grounding
Project: Aug. 2026 – Present; manuscript under review

- Developed a dataset and benchmark that ground news images in shared cross-document event structures, supporting analysis of which subevents outlets foreground and how images represent them.
- Built LLM-assisted extraction, consolidation, and verification workflows that integrate event information across articles while retaining source quotations and checking evidential support for structured representations.
- Evaluated language and vision-language models through controlled comparisons of context, event structure, and stepwise inference, separating the effects of evidence representation from the decisions models make with it.
- Developed annotation guidelines and trained annotators to assess image–event relations, translating research concepts into practical criteria and resolving ambiguous cases to support consistent human evaluation.

Selected Research Projects

Context Selection and LLM Fine-Tuning on Graphs Published 2025
Worcester Polytechnic Institute *KGBigdata, IEEE Big Data*

- Investigated how much neighboring context LLMs need for node classification, varying the amount and composition of graph context during training and inference to study its effect on predictive performance.
- Developed a multi-model text augmentation framework using summaries from five language models, combining selected graph context with original and augmented text to fine-tune Llama-3.1-8B.
- Achieved **74.31% accuracy on ogbn-arxiv** and **85.15% on ogbn-products** using rsLoRA and limited neighbor information, outperforming the reported comparison baselines in a study covering 11 methods.

Detecting AI-Generated Meme Ideas Dec. 2025 – Jul. 2026
Task design and evaluation of generated content *AACL-IJCNLP 2026 SRW*

- Designed a task comparing human and AI-generated meme ideas using shared visual templates, building a dataset from 2,735 Reddit memes through VLM extraction and template-matched generation.
- Benchmarked four LLMs across nine generator–detector pairs with zero-shot, few-shot, and chain-of-thought prompting, conducting five seeded runs per setting to compare detection behavior across generators.
- Fine-tuned Gemma 4 with LoRA on 1,000 examples and evaluated simpler supervised baselines, keeping human and generated ideas from each meme in the same data split.
- Combined human evaluation and lexical analysis to identify recurring workplace and technology themes in AI-generated ideas, revealing predictable content patterns that help explain detection performance.

Teaching Experience

Teaching Assistant, Brandeis University Aug. 2025 – Sep. 2026
Introduction to Machine Learning (Spring 2026) *Waltham, MA*
Fundamentals of Natural Language Processing I (Fall 2025)

Teaching Assistant, Worcester Polytechnic Institute Aug. 2024 – May 2025
Data Visualization (Spring 2025) *Worcester, MA*
Introduction to Algorithms: Design and Analysis (Fall 2024)

Technical Skills

Programming: Python, SQL, R, C/C++.
ML/NLP: PyTorch, Transformers, vLLM, PEFT (LoRA/rsLoRA), Unsloth, PyTorch Geometric, LangChain.
Data and analysis: scikit-learn, pandas, NumPy, spaCy, Matplotlib, Scattertext.

Academic Service

Reviewing contributions: SIGIR 2024, SIGIR 2025, WSDM 2025.